MIL-D-45503A
6 DECEMBER 1961
SUPERSEDING
MIL-D-45503
22 MAY 1959

MILITARY SPECIFICATION

DRIVE, HYDRAULIC: M12

This specification has been approved by the Department of Defense and is mandatory for use by the Department of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 This specification covers one type of hydraulic drive for the gun, 20-mm, Automatic: M61. The drive is composed of a bail piston motor and a drive unit which transmits torque to the gun through a planetary gearing system.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-B-585 — Boxes, Wood, Wirebound.

PPP-B-591 — Boxes, Fiberboard, Wood-Cleated.

PPP-B-601 — Boxes, Wood, Cleated. Plywood.

PPP-B–621 — Boxes, Wood, Nailed and Lock-Corner.

PPP-B-636 — Box, Fiberboard.

PPP-B-640 — Boxes, Fiberboard, Corrugated, Triple, Wail.

MILITARY

MIL-P-116 — Preservation, Methods of.

MIL-H-5606 — Hydraulic Fluid, Petroleum Base, Aircraft and Ordnance.

MIL-Q-9858 — QuaIity Control System Requirements.

MIL-W-13855 — Weapons, Small Arms, General Specification for.

MIL-I-45607 — Inspection Equipment, Supply and Maintenance of.

MIL-C-45662 — Calibration of Standards.

FSC 1005

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STANDARDS

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-109 — Inspection Terms and Definitions.

MIL-STD-643 — Evaluation of Contractor Quality Control Systems.

DRAWINGS

ORDNANCE CORPS

J7269993 — Drive, Hydraulic, M12.

IEL7274581 — Index of Inspection Equipment Lists.

PUBLICATIONS

ORDNANCE CORPS

LPDS-7269993 — List of Packaging Data Sheets.

LSQAP-7269993 — Index of Supplementary Quality Assurance Provisions.

MIL-D-45503 P7269993 — Packaging Data Sheet for Drive, Hydraulic, M12.

SQAP-7269993 — Supplementary Quality Assurance Provisions for Drive, Hydraulic, M12.

(Copies of specifications, standards, drawings, and publications required by Contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 **Production samples.** On each contract, three drives and samples of parts representing initial production shall be taken by the inspector and shall be forwarded free on board (f.o.b.) carrier's equipment, wharf, or freight station at a specified city or shipping point at or near Contractor's plant to a testing agency designated by the procuring activity (see 6.1). Samples shall be representative of materiel which has been inspected and determined to be acceptable by the inspector. The production samples of drives shall be contained in the pilot pack (see 5.1).

3.2 **Materials and construction.** Drives and parts shall conform to the materials and construction requirements specified herein and on drawings referenced on drawing J7269993, and shall be in accordance with the applicable provisions of Specification MIL-W-13855.

3.3 **Design.** Drives and parts shall conform to the design specified herein and on drawings referenced on drawing J7269993, and shall be in accordance with the applicable provisions of Specification MIL-W-13855.

3.3.1 *Bolts, nuts, and screws.* Bolts, nuts, and screws shall be tight and shall fasten parts securely without relative movement as evidenced by manual examination.

3.3.2 *Direction of rotation.* Rotation of the output shaft spline shall be clockwise (viewing from drive output) with the M1 port as the pressure inlet.

3.3.3 *Output shaft spline.* The output shaft spline shall be fastened securely to the output shaft assembly by the nut without relative movement as evidenced by manual examination.

3.3.4 *Safety wiring.* Safety wiring shall be in accordance with the applicable drawings and shall prevent loosening of parts as evidenced by visual and manual examination.

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3.3.5 *Temperature*. Drives shall withstand a temperature range extending from -65° ± 3.6°F. to 275° ± 3.6°F. when tested as specified in 4.4.2.

3.3.6 *Torque.*

3.3.6.1 *Running torque*. The torque output shall be not less than 2,000 pound-inches, the flow from M13 (drain port) shall be not more than 2.0 gallons per minute (gpm), and the torque versus speed and torque versus pressure shall be in accordance with the requirements of figures 1 and 2 respectively when the drive is tested as specified in 4.4.3.1.

3.3.6.2 *Stalled torque.* The torque output shall be not less than 1,930 pound-inches and the flow from M2 (return port) and M3 shall be not more than 2.0 gpm respectively when the drive is tested as specified in 4.4.3.2.

3.3.7 *Functioning.* Drives shall operate under load without storage or unserviceable parts attributable to the drives and shall meet the performance requirements specified in table I when tested as specified in 4.4.4. The steady state rate shall be calculated using the last 50 percent of the duty cycle, excluding the stopping time.

TABLE I. *Performance requirement.*

Acceleration time (maximum)	Steady state rate (average)	Stopping time (maximum)
0.4 seconds (from start of drive rotation to minimum steady state rate).	1,000 to 1,200 revolutions per minute (rpm).	0.4 seconds (from minimum steady state rate to complete stop).

3.3.8 *Interchangeability.* All parts, except component parts of the motor, shall be interchangeable. (In normal assembly operations there shall be no objections interposed to preferential assembly of parts provided that all parts are dimensionally acceptable.) Drives and repair parts shall be capable of meeting the interchangeability tests specified in 4.3.3 .1.2.3 and 4.3.3.2.1.

3.3.9 *Reliability.* Drives shall be capable of passing a reliability test of 300 duty cycles without functioning failures (see 3.3.7) when tested as specified in 4.4.6.

3.3.10 *Leakage.* There shall be no external leakage of hydraulic fluid except that leakage from the drive shaft seal shall be allowed in accordance with table II when the drive is tested as specified in 4.4.7.

3.4 **Historical record.** Upon completion of all inspection, required data shall be entered on DD Form 829, Historical Record for Aeronautical Equipment, and one completed form shall be provided with each drive.

TABLE II. *Permissible leakage.*

Drive condition	Test condition	Rate of leakage (maximum)
After torque and functioning tests	static	one drop in 30 minutes
	dynamic.	one drop in 5 minutes
After reliability test	static	one drop in 2 minutes
	dynamic.	100 drops in one hour

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3.5 **Marking.** Drives and parts shall be clearly marked in accordance with the applicable drawings and Specification MIL-W-13855. The identification plate shall be fastened securely to the drive by the screws without relative movement as evidenced by manual examination.

3.6 **Workmanship.** Workmanship shall be in accordance with the workmanship requirements of Specification MIL-W-13855.

4. QUALITY ASSURANCE PROVISIONS

4.1 **General quality assurance provisions.** The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept com plete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 *Contractor's quality control system.* The contractor shall provide and maintain a quality control system in accordance with Specification MIL-Q-9858 and Standard MIL-STD-643.

4.1.2 *Government verification.* All quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of (a) surveillance of the operations to determine that the practices, methods, and procedures of the written inspection plan are being properly applied, and (b) Government product inspection to measure quality of product offered for acceptance. Deviation from the prescribed or agreed-upon procedures, or instances of poor practices which might have an effect upon the quality of the product will be immediately called to the attention of the contractor. Failure of the contractor to promptly correct deficiencies discovered shall be cause for suspension of acceptance until correction has been made or until conformance of product to prescribed criteria has been demonstrated. To avoid interference with operations, the contractor shall designate a responsible official or officials to whom the inspector wiIl report such instances. Where there is satisfactory evidence of high quality of the product which is a definite result of the contractor's quality control system, the amount of Government examination may be adjusted to a minimum.

4.1.3 *Inspection terms and definitions.* Inspection terms and definitions used herein are in accordance with Standard MIL-STD-109, Whenever the term "inspector" is used herein, it denotes Government inspector.

4.2 **Production sample inspection.** Inspection of production samples (see 3.1) and pilot packages (see 5.1) shall be performed at the designated testing agency. Production samples and pilot packages shall be subjected to the inspection specified herein and such other inspection as is necessary to determine compliance with the requirements of the contract.

4.3 **Inspection provisions.**

4.3.1 *Inspection lot.* The formation, size, and presentation of inspection lots of drives, parts, and packaging shall be in accordance with Standard M1L-STD-105. The inspector shaIl insure that inspection lots are as large as practicable, in consideration of quality history, manufacturing conditions, and Contractor's delivery schedule, and within the limitations of Specification MIL-W-13855. Drives shall be assembled from lots of component parts that have met all inspection requirements specified herein. Reliability test lot size shall be as specified in 4.3.3.1.2.4.1.

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4.3.2 *Examination.*

4.3.2.1 *Drives.*

4.3.2.1.1 Drives shall be examined for completeness of manufacture, assembly, finish, marking, and workmanship in accordance with Supplementary Quality Assurance Provisions (SQAP) SQAP-7269993.

4.3.2.1.2 Drives which have passed all examinations after successful completion of testing shall be stamped by the inspector with the Department of Defense complete inspection approval stamp as specified on the applicable drawing.

4.3.2.2 Component parts and concurrent repair parts. Examination of parts shall be performed in accordance with the applicable SQAP'S listed on the Index of Supplementary Quality Assurance Provisions LSQAP-7269993 (see 6.2). The Contractor's examination of parts shall be accomplished prior to their assembly into the end item or submission for acceptance as repair parts.

4.3.2.3 *Packaging.* Examination of packaging of drives shall be performed in accordance with the classification of defects and acceptable quality levels (AQL's) specified in 4.3.2.3.1 (see 6.1). Sample size shall be in accordance with Standard MIL-STD-105, using inspection level I. The following provisions shall apply:

(a) The AQL's are specified as percent defective.

(b) An individual AQL is specified for each listed defect, not for a group of defects.

(c) Examination for packaging defects specified in 4.3.2.3.1 shall apply to each item of the applicable sample of drives, unit packages, or exterior pack, as applicable.

4.3.2.3.1 *Classification of defects for packaging.* Unless otherwise specified in each listed defect, the packaging requirements are specified on Packaging Data Sheet MIL-D-45503 P7269993.

Categories	*Defect*	*AQL*
Critical:	None defined.	
Major:		
101	Illegible or incorrect marking	1.0
102	Improper level of packaging and packing (see procurement documents) .	1.0
103	Inadequate cleaning and drying . . .	1.6
104	Improper preservative application . .	1.5
105	Missing or loose shipping plugs	1.6
106	Missing DD Form 829	1.6
107	Improper assembly of packages	1.5
108	Improper closure of unit packages . .	1.5
109	Improper closure and strapping of shipping containers	1.5
Minor:		
201	Workmanship (see 5.4)	4.0

4.3.2.3.2 Examination of packaging of repair parts shall be performed in accordance with the applicable SQAP'S.

4.3.3 *Testing.*

4.3.3.1 *Drives.*

4.3.3.1.1 *Production sample testing.* One of the sample drives, forwarded in accordance with 3.1, shall be subjected to the temperature tests by the Government at a testing agency designated by the procuring activity (see 6.1). The drive shall be subjected to the examination specified in 4.3.2.1.1 and the torque and functioning tests specified in 4.4.3 and 4.4.4 before being subjected to the temperature tests specified in 4.4.2. Failure of the drive to meet the requirements shall result in rejection of the production sample, subject to retest at the discretion of the procuring activity.

4.3.3.1.2 *Acceptance testing.*

4.3.3.1.2.1 *Torque and functioning testing.* The contractor shall test each drive for torque and functioning using the test methods specified in 4.4.3 and 4.4.4 respectively. Drives failing to meet the requirements shall be rejected.

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4.3.3.1.2.2 *Leakage testing.* The Contractor shall test each drive for leakage after the torque and functioning tests, and shall test each reliability tested drive for leakage using the test method specified in 4.4.7. Any drive failing to meet the requirements during 100 percent testing shall be rejected; and failure of any reliability tested drive to meet the requirements shall result in rejection of the represented lot.

4.3.3.1.2.3 *Interchangeability testing.* The inspector shall subject a sample of three drives from each inspection lot to interchangeability testing using the test method specified in 4.4.5.1. Drives taken for interchangeability testing shall have been found satisfactory in all other examinations and tests. Test frequency may be reduced to not less than one test of three drives each month when a record of consistently satisfactory results has been established. The three drives shall be inspected for and shall comply with the requirements for torque and functioning before and after interchange of parts using the test methods specified in 4.4.3 and 4.4.4 respectively. Hand refinement of parts will be allowed on not more than one drive during interchanging of parts provided that no part is altered beyond drawing requirements. Failure of drives to meet the requirements of the interchangeability test or failure of any drive to meet the requirements of the torque and functioning test shall be cause for retest or rejection of the represented lot. At the discretion of the inspector, an interchangeability retest may be allowed without reconditioning the lot of drives. Failure in the retest shall cause rejection of the represented lot subject to reconditioning and further test as a reconditioned lot. A sample of six drives from each retest or reconditioned lot shall be tested using the same procedure described above.

4.3.3.1.2.4 *Reliability testing.*

4.3.3.1.2.4.1 *Lot size.* The initial reliability test lot size shall consist of the initial month's production; subsequent reliability test lot sizes shall consist of 100 drives.

4.3.3.1.2.4.2 *Procedure.* One drive selected by the inspector from each reliability lot shall be tested by the Contractor for reliability using the test method specified in 4.4.6. If the reliability requirements are not met, the represented lot shall be rejected subject to retest or reconditioning and further test as a reconditioned lot. A reliability retest of another drive from the same lot shall be made without reconditioning the represented lot, unless in the opinion of the inspector the failure indicate serious defects in the item, in which case retest shall be made only if authorized by the procuring agency. Failure of the drive in the retest to meet the requirements shall cause rejection of the represented lot subject to reconditioning and further testing as a reconditioned lot. Prior to submission of a lot of drives as a reconditioned lot, the cause of failure shall be determined and contractor correction shall be effected on all drives in the lot. Sample size and test methods for reconditioned lots shall be the same as for retest.

4.3.3.2 *Component parts and concurrent repair parts testing.* Raw material testing, part testing, and certification shall be performed in accordance with the applicable SQAP'S. This will include chemical analysis and physical tests of materials, and tests of protective finish, heat treatment, bonding, and function of parts as applicable. The Contractor shall accomplish these tests prior to assembly of parts into the end item.

4.3.3.2.1 *Concurrent repair parts interchangeability testing.* The inspector shall subject at least 2 parts from each inspection lot of concurrent repair parts to the interchangeability test specified in 4.4.5.2.

4.3.3.3 *Packaging.*

4.3.3.3.1 *Drives.*

4.3.3.3.1.1 The Contractor shall furnish the inspector with certification that the packaging materials conform to the applicable packaging data sheets and specifications.

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4.3.3.3.1.2 *Determination of cleanliness testing.* The Contractor shall test items from each inspection lot for determination of cleanliness using the test method specified in 4.4.8.1. Sampling shall be in accordance with Specification MIL-P-116.

4.3.3 .3.1.3 *Quick leak testing.* The Contractor shall test interior packages from each inspection lot for quick leak using the test method specified in 4.4.8.2. Sampling shall be in accordance with Specification MIL-P–116.

4.3.3.3.2 *Repair parts.* Testing of packaging of repair parts shall be performed in accordance with the applicable SQAP's.

4.2.4 *Inspection equipment.*

4.3.4.1 General measuring and testing equipment provisions are contained in Specification MIL-Q-9858. Responsibilities for acquisition, maintenance, and disposition of acceptance inspection equipment prescribed on lists contained on the Index of Inspection Equipment Lists, Drawing IEL727458L shall be in accordance with Specification MIL-I-45607 and MIL-C-45662 (see 6.1).

4.3.4.2 The inspector will determine that the Contractor has available and utilizes correctly, gaging, measuring, and test equipment of required accuracy and precision and that the instruments are of proper type and range to make measurements of the required accuracy. The calibration of gages, standards, and instruments will be periodically checked by authorized Government personnel.

4.3.4.3 Unless otherwise specified in procurement documents, drawings of Contractor designed inspection equipment shall be forwarded to Springfield Armory, Springfield 1, Mass., Attn: ORDBD-EG (see 4.4.1.3, 4.4.3.3, 4.4.4, and 6.1).

4.4 **Test methods.**

4.4.1 *Test conditions.* Unless otherwise specified in the individual test methods, the following test conditions shall apply.

4.4.1.1 *Hydraulic fluid.* The hydraulic fluid shall conform to Specification MIL-H-5606.

4.4.1.2 *Temperature.* The hydraulic fluid inlet temperature shall be 130° & 10°F. and the temperature differential between the drive and the hydraulic fluid shall not be more than 100°F.

4.4.1.3 *Power supply and control equipment.* The power supply and control equipment shall be capable of maintaining a differential pressure of 2,800 pounds per square inch (psi) across the drive (M1 to M2) and a flow of more than 23.8 gpm to the drive continuously throughout the cycling period of the drive. In addition, the control equipment shall prevent variations of more than one percent in the steady state rate once a steady state rate has been reached, and shall provide a braking device to stop the drive and load within the time limit specified in table I. The M2 and M3 pressure shall not exceed 200 psi respectively. Aircraft installation equipment or Government approved equipment of the Contractor's design which simulates the aircraft installation equipment shall be used for applying the power supply to the drive (see 4.3.4.3).

4.4.2 *Temperature tests.* The drive shall be tested for temperature requirements (see 3.3.5) as specified in 4.4.2.1 and 4.4.2.2.

4.4.2.1 *Low temperature.* The drive shall be placed in a test chamber and conditioned at a temperature of -65° ± 3.6°F. for 5 hours. After conditioning the drive and while at this temperature, hydraulic fluid (see 4.4.1.1) shall be circulated through the M4 drive port for 10 minutes at a rate of 0.70 + 0.02 gpm. The hydraulic fluid inlet temperature shall be maintained at 130° + 10°F. during this circulation period. Immediately after this 10 minute period, the drive shall be operated for 11 seconds at any steady state rate between 1,000 to 1,200 rpm using

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a variable load and the power supply and control equipment specified in 4.4.1.3. The differential pressure across the drive shall be not more than 2,800 psi and the flow required by M1 shall be not more than 23.8 gpm. At conclusion of this operation period, the drive shall be tested for the torque and functioning requirements using the test methods specified in 4.4.3 and 4.4.4 respectively.

4.4.2.2 *High temperature.* The test method for the high temperature test shall be the same as specified in 4.4.2.1 except that the drive shall be conditioned at a temperature of 275° ± 3.6°F. with a relative humidity of 15 plus 5 percent minus 0 percent and the hydraulic fluid inlet temperature shall be maintained at 75° ± 5°F.

4.4.3 *Torque tests.*

4.4.3.1 *Running torque.* The drives shall be tested for running torque requirement (see 3.3.6.1) using the test conditions specified in 4.4.1 and a variable load to hold the drive speed at 1,050 rpm. The differential pressure across the drive shall be not more than 2,800 psi and the flow required by M1 shall be not more than 23.8 gpm. The drive torque and the hydraulic fluid pressure and flow to M1 and from M2 and M3 shall be measured and recorded for each test. In addition, the drives shall be tested for torque versus speed and torque versus pressure requirement (see 3.3.6.1) using the test conditions specified in 4.4.1 and a variable load. The torque versus speed and torque versus pressure curves shall be taken and recorded for each test. The accuracy to which the torque shall be determined shall be within 2 percent.

4.4.3.2 *Stalled torque.* The drives shall be tested for stalled torque requirement (see 3.3.6.2) using the test conditions specified in 4.4.1 and with the drive shaft spline locked. The differential pressure across the drive shall be not more than 2,800 psi. The drive torque output and the hydraulic fluid pressure and flow to M1 and from M2 and M3 shall be measured and recorded for each test. The accuracy to which the torque shall be determined shall be within 2 percent.

4.4.3.3 Government approved inspection equipment of the Contractor's design shall be used (see 4.3.4.3).

4.4.4 *Functioning test.* The drives. shall be tested for functioning requirement (see 3.3.7) at least once through its function cycle with the power on for 2 seconds assembled to a slave M61 gun using a belt of dummy cartridges and the actuator booster assembly loaded to simulate booster function conditions or using Government approved test equipment of the Contractor's design which simulates the gun conditions (see 4.3.4.3). The differential pressure across the drive shall be not more than 2,800 psi and the flow required by Ml shall be not more than 23.8 gpm. Other test conditions shall be as specified in 4.4.1. The acceleration time, steady state rate, and stopping time shall be measured and recorded for each drive.

4.4.5 *Interchange of parts.*

4.4.5.1 *Drives.* Drives shall be tested for interchange of parts (see 3.3.8) by disassembling and then reassembling parts using the parts and prearranged system prescribed in SQAP-7269993.

4.4.5.2 *Concurrent repair parts.* Concurrent repair parts shall be tested for interchangeability requirement (see 3.3.8) by disassembling two drives, previously tested in 4.3.3.1.2.3, as necessary and then reassembling them using the concurrent repair parts. No hand refinement of parts will be allowed, and the drives shall operate and function properly. This test may be performed independently of the drive interchangeability test specified in 4.3.3.1.2.3, and at more frequent intervals using accepted drives taken from current production.

4.4.6 *Reliability test.*

4.4.6.1 The drives shall be tested for reliability requirement (see 3.3.9) for 300 duty

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cycles in series of 5 duty cycles each assembled to the load specified in 4.4.4. The differential pressure across the drive shall be not more than 2,800 psi and the flow required by M1 shall be not more than 23.8 gpm. A duty cycle shall be defined as time on for 2 seconds and time off for 5 seconds. The acceleration time, steady state rate, and stopping time shall be measured and recorded for the last duty cycle of each fourth series. Cooling shall be after each series. Cleaning and lubricating of test equipment shall be allowed after each two series and at the end of each day's testing. No cleaning, lubricating, or disassembly of the drives shall be allowed. Other test conditions shall be as specified in 4.4.1.

4.4.6.2 At the completion of the realizability test, drives shall be disposed of as specified in the contract (see 6.1).

4.4.7 *Leakage test.* The drives shall be tested for leakage (see 3.3.10) by application of 45 psi ± 5 psi motor case pressure when testing for leakage in the static condition, and by application of 200 psi ± 15 psi motor case pressure when testing for leakage in the dynamic condition. When testing the drive for leakage in the static condition, the drive shall be completely filled with hydraulic fluid; and when testing the drive for leakage in the dynamic condition, the drive shall be operated with the load specified in 4.4.4 using the test conditions specified in 4.4.1.

4.4.8 *Packaging tests.*

4.4.8.1 *Determination of cleanliness.* The applicable surfaces of each sample unit shall be subjected to the determination of cleanliness test in accordance with Specification MIL-P-116, except that the wipe test shall apply only to the needle bearing race and the output shaft spline.

4.4.8.2 *Quick leak.* The level A sample unit packages shall be subjected to the quick leak test specified in Specification MIL-P-116.

5. PREPARATION FOR DELIVERY

5.1 **Pilot pack.** On each contract a pilot pack consisting of a complete and packed shipping container representing initial production shall be forwarded in accordance with 3.1. The pilot pack shall consist of 3 unit packages containing drives.

5.2 **Packaging data sheets.** Quantity of drives per unit package shall be as specified under "Logistic Data" on Packaging Data Sheet MIL-D-45503 P7269993. All other logistic data are for informational use only. Quantities of drives indicated for the exterior packs on the packaging data sheets are maximum, and the size of exterior packs are for this maximum quantity.

5.3 **Preservation, packaging, packing, and marking.** Drives shall be preserved, unit packaged, packed, and marked in accordance with the requirements of Packaging Data Sheet MIL-D-45003 P7269993.

5.4 **Workmanship.** Adequate controls shall be utilized to check for contamination of cleaning solvents and preservatives.

5.5 **Repair parts.** Repair parts shall be prepared for delivery in accordance with the applicable packaging data sheets listed on the List of Packaging Data Sheets, LPDS-7269993.

5.5.1 *Packaging data sheets.* Quantity of items per unit and intermediate package shall be as specified under "Logistic Data". All other logistic data are for informational uses only. Quantities of repair parts indicated for the exterior packs in the packaging data sheets are maximum, and the sizes of exterior packs indicated are for this maximum quantity. When repair parts are to be shipped in quantities less than the maximum quantity specified on the packaging data sheets, multipacks of various repair parts shall be required. Multipacks for level A repair parts shipment shall be within a maximum gross weight limitation of 200 pounds

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plus 10 percent, with shipping containers conforming to specifications below:

PPP-B-585 — Class 3.

PPP-B-591 — Style A, overseas type

PPP-B-601 — Style A, overseas type

PPP–B–621 — Class 2, style 4.

PPP-B-640 — C1ass 2.

When the total multipack shipment volume is less than 3 cubic feet, or the shipment weight is less than 70 pounds, a fiberboard box conforming to class 2 of Specification PPP-B–636 shall be used for level A shipments.

5.5.2 Multipacks for levels B and C repair parts shipments shall be in lightweight economical containers capable of making a single trip shipment without damage to the contents and shall be acceptable by common carrier at the lowest rate.

6. NOTES

6.1 **Ordering data.** Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Lists of drawings and specifications pertinent to the drive, showing applicable revision dates.

(c) Index of supplementary quality assurance provisions pertinent to the drive, showing applicable revision dates.

(d) Index of inspection equipment lists pertinent to the drive, showing applicable revision dates.

(e) Number of production samples required for tests (see 3.1).

(f) That packages opened for examination shall be repackaged by the contractor at the Contractor's expense.

(g) Disposition of reliability tested drives (see 4.4.6.2).

(h) Shipping instructions for production samples and pilot pack (see 3.1 and 5.1).

(i) Shipping instructions for drawings of Contractor designed test equipment if other than specified (see 4.3.4.3).

(j) Selection of applicable levels of preservation, packaging, and packing.

(k) Packaging instructions for repair parts (see 5.5).

(l) Place of final inspection and acceptance (see "Notes" in Specification MIL-W-13855).

(m) List of acceptance inspection equipment to be furnished the Contractor (see 4.3.4.1) and responsibilities for other Government property to be furnished the Contractor.

(n) Responsibilities for furnishing dummy cartridges and links.

(o) Quantity, shipping instructions, and test procedures for drives and parts when an interplant interchangeability test is required.

(p) Procedures and methods for demilitarizing and disposing of rejected material.

(q) Disposition of Government furnished property.

(r) When action by a testing agency is required, work programing will be effected with the testing agency at the earliest practicable date.

6.2 Supplementary Quality Assurance Provisions (SQAP'S) may be obtained from Springfield Armory, Springfield 1, Mass. ATTN: ORDBD-EG.

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Custodians:

Army-Ord
Air Force-WRA

Preparing activity:

Army--Oral

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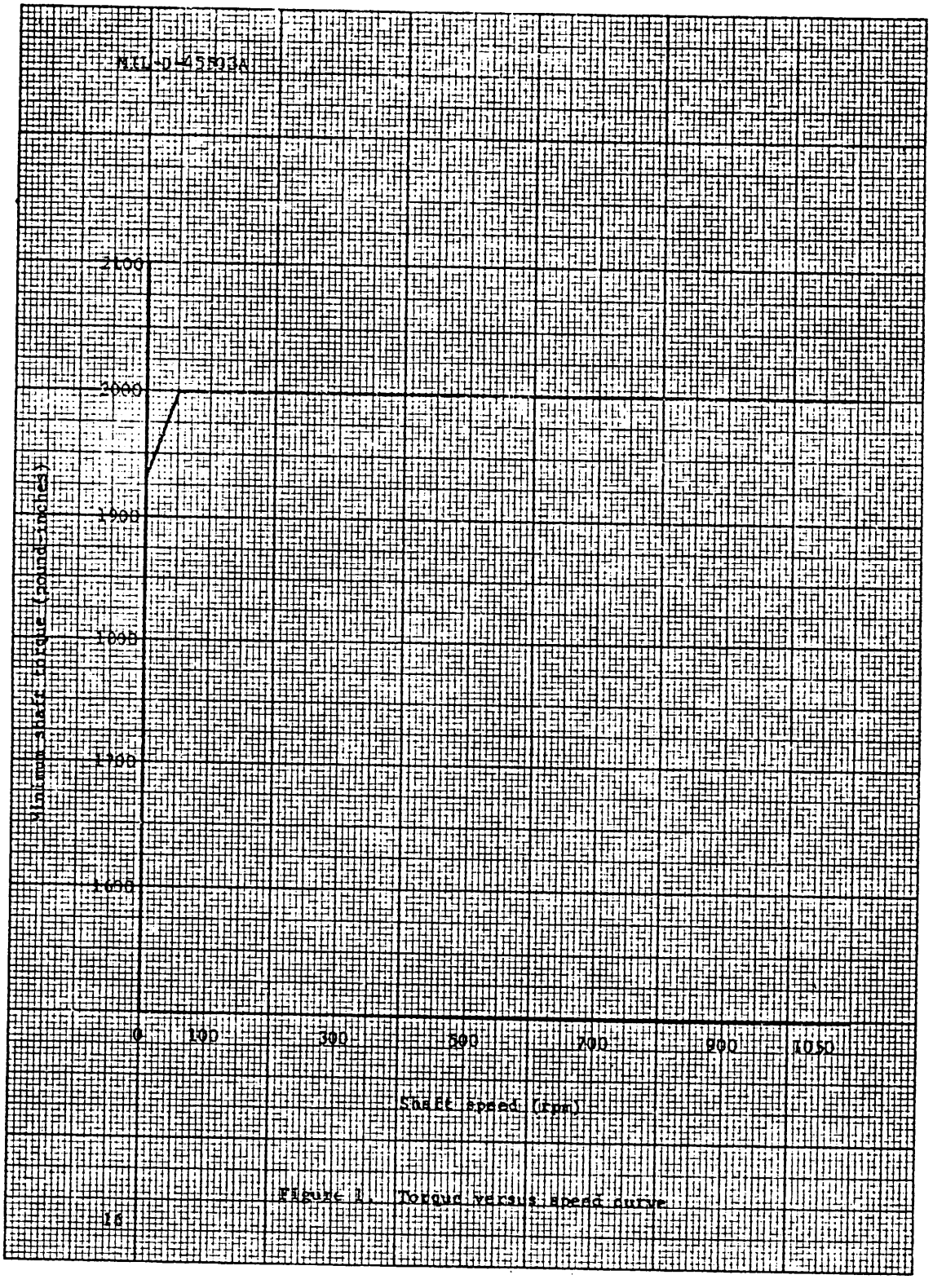


FIGURE 1. *Torque versus speed curve - shaft speed (rpm).*

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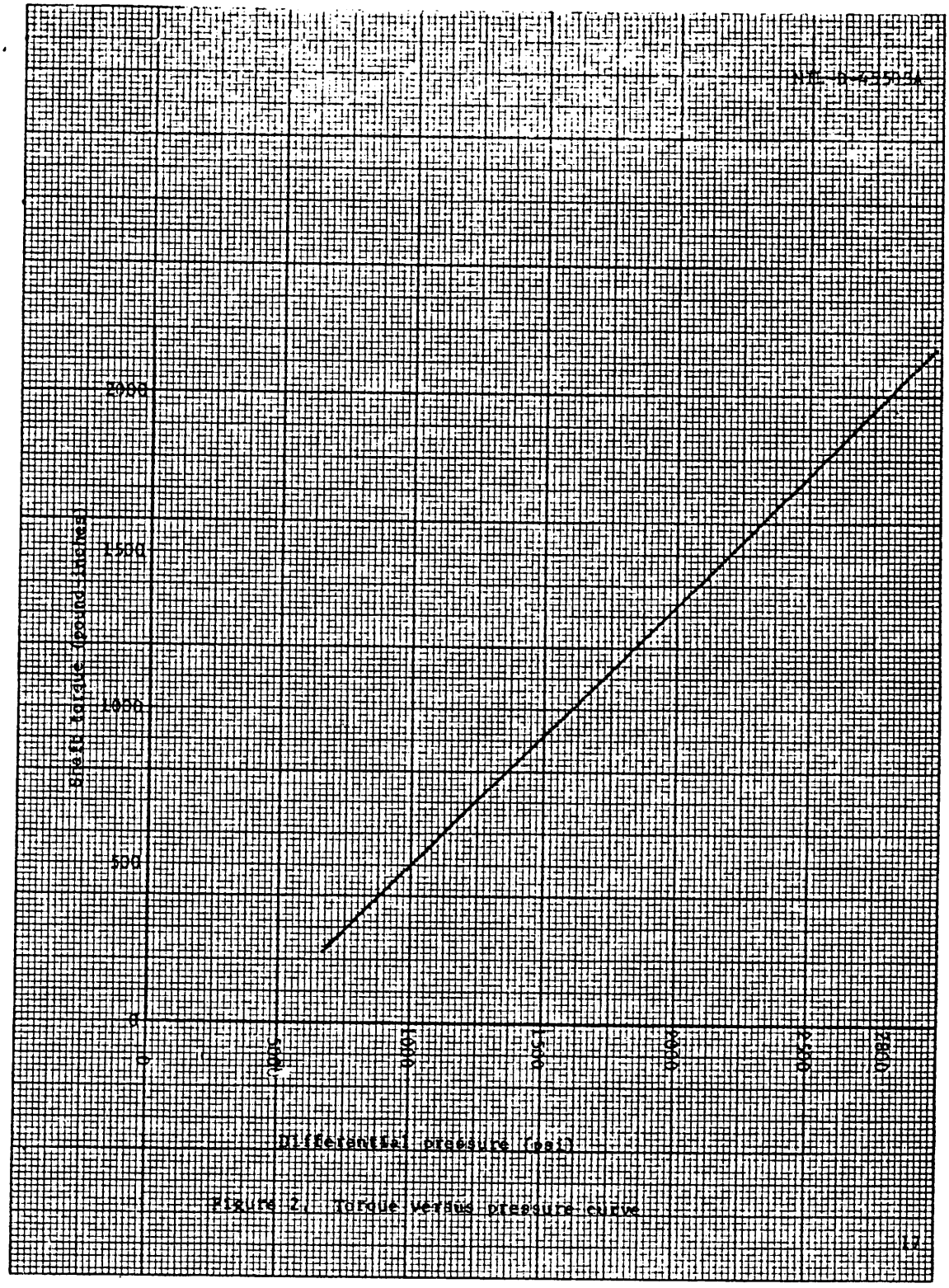


FIGURE 2. *Torque pressure curve - differential pressure (psi).*

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NOTICE
OF VALIDATION

MIL-D-45503A
NOTICE 1
28 May 1987

MILITARY SPECIFICATION

DRIVE, HYDRAULIC: T46

MIL-D-45503A, dated 6 December 1961, has been reviewed and determined to be valid for use in acquisition.

Custodians:
Army - AR
Air Force -99

Preparing activity:
Army - AR

AMSC N/A

FSC 1005

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